DSA-210 Final Report

Gender Bias in Online Toxicity Analysis

Project Summary

This project investigates whether comments containing female-related expressions exhibit different toxicity levels compared to general online comments. The analysis combines exploratory data analysis, hypothesis testing, and machine learning methods.

1. Motivation

Online communication platforms contain millions of comments, discussions, and interactions every day. However, these platforms also contain toxic language, harassment, and hate speech. Discussions about gender bias and misogyny have become increasingly common in digital environments, especially on social media platforms and public forums. The purpose of this project is to investigate whether comments containing female-related expressions exhibit different toxicity patterns compared to general online comments. This topic is both socially relevant and technically interesting because it combines natural language processing, statistical analysis, and machine learning. Rather than focusing only on prediction performance, the project aims to construct a complete data science pipeline involving preprocessing, feature engineering, exploratory analysis, statistical testing, visualization, and machine learning classification.

2. Dataset

The project uses the publicly available Jigsaw Toxic Comment Classification Dataset from Kaggle. The dataset contains online comments labeled with multiple toxicity categories including: • toxic • severe_toxic • obscene • threat • insult • identity_hate This dataset is commonly used in NLP and toxicity research because it provides a large-scale collection of labeled online comments suitable for both statistical analysis and supervised machine learning.

3. Data Preparation and Feature Engineering

Several preprocessing operations were applied before the analysis stage. First, comment texts were converted to lowercase in order to standardize the text format. A keyword-based filtering approach was then used to identify comments potentially related to women. The selected keywords included: • she • woman • women • female • girl Using these expressions, a binary feature called women_related was created. A combined toxicity score was also calculated using the average of the toxicity-related labels. This score was used throughout the project as a simplified numerical representation of overall toxicity.

4. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to examine the structure of the dataset and compare toxicity distributions between groups. Several visualizations were created: • Comment group count comparison • Average toxicity score comparison • Toxicity score histogram • Toxicity score boxplot The histogram visualizations showed that most comments have low toxicity scores, while a smaller number of comments contain extremely toxic content. This produced a skewed distribution with visible outliers. The boxplots also revealed differences in variability between women-related comments and other comments.

5. Statistical Analysis

To move beyond visual inspection, multiple statistical tests were applied. Before applying parametric tests, normality assumptions were checked using the Shapiro-Wilk normality test. The results indicated that the toxicity distributions were not perfectly normal. The following statistical methods were applied: • Independent t-test • Mann-Whitney U test Both tests suggested statistically significant differences between women-related comments and other comments.

6. Machine Learning Methods

Machine learning methods were applied to classify toxic comments. The selected features included: • women_related • toxicity_score The dataset was divided into training and testing sets using an 80/20 split. Two machine learning models were implemented: • Logistic Regression • Decision Tree Classifier The models were evaluated using: • Accuracy score • Classification report • Confusion matrix

7. Results and Findings

The analysis suggests that comments containing female-related expressions exhibit different toxicity distributions compared to other comments. The statistical tests indicated significant differences between groups, and the machine learning models achieved reasonable performance in predicting toxic comments. Logistic Regression provided a simple and interpretable baseline model, while the Decision Tree model captured more flexible nonlinear relationships.

8. Discussion

The project demonstrates how statistical analysis and machine learning techniques can be combined to investigate social patterns in online communication. One important limitation is the simplicity of the keyword-based filtering method. Some comments may contain female-related words without targeting women directly, while other comments may imply gender-related content without explicit keywords. In addition, toxicity is context-dependent, and simple numerical representations cannot fully capture sarcasm, irony, or conversational nuance.

9. Limitations and Future Work

This project has several limitations. The keyword list is relatively small and may not fully represent gender-related language. Furthermore, the toxicity score simplifies multiple toxicity categories into a single numerical variable. Future improvements may include: • Transformer-based NLP models such as BERT • Larger and more diverse datasets •

Context-aware language analysis • More advanced text embeddings • Hyperparameter optimization

10. Conclusion

This project investigated whether comments containing female-related expressions exhibit different toxicity patterns compared to general online comments. The project successfully combined preprocessing, feature engineering, exploratory analysis, statistical testing, and machine learning methods within a single workflow. The findings suggest that women-related comments may exhibit distinct toxicity distributions and demonstrate how data science techniques can be applied to socially relevant online communication problems.

11. AI Usage Disclosure

AI tools were used during the project for debugging support, improving report structure, refining wording, and organizing explanations. However, preprocessing decisions, notebook execution, statistical analysis, machine learning implementation, and interpretation of results were completed and reviewed by the author.

Technologies Used

Technology	**Purpose**
Python	Programming Language
Pandas	Data Processing
Matplotlib	Visualization
SciPy	Statistical Testing
Scikit-learn	Machine Learning
Jupyter Notebook	Interactive Analysis

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